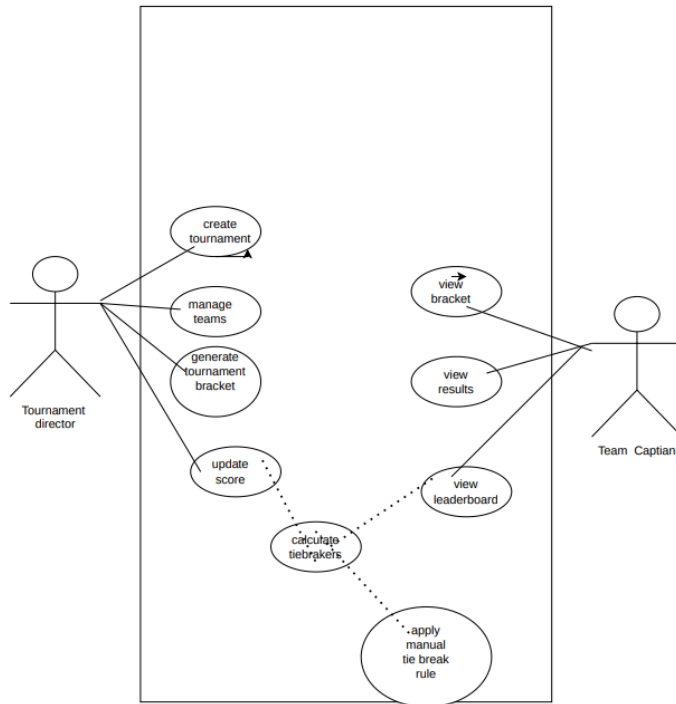


FUNCTIONAL AND NON FUNCTIONAL REQUIREMENTS

REQ_ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALIZE
FR-001	FUNCTIONAL	The system shall automatically generate seeded single-elimination and double-elimination tournament brackets based on the initial team rankings.	High	Pass: After rankings and tournament type are selected, a valid bracket is generated and teams are placed according to their seeds. Fail: Invalid or duplicate match pairings are generated.	Core functionality required to organize the tournament.
FR-002	FUNCTIONAL	The System should allow tournament director to add participant and select the tournament format	high	Pass: the Tournament director should be able to add or edit the teams	Required to organize the Tournament
FR-003	FUNCTIONAL	The system Should allow the Tournament Director to record and update match scores for completed or live matches.	high	Pass: When scores are submitted, the system saves and displays the updated scores	Required for tracking tournament progress
FR-004	FUNCTIONAL	The system should automatically advance the winning team to the next match	high	Pass: After a winner is determined, the team is automatically placed in the correct next bracket position Fail: The winning team is placed incorrectly or does not advance.	Ensures the tournament bracket progresses correctly without manual updates.

FR--005	FUNCTIONAL	The system shall calculate tiebreakers and update the league standings based on submitted match scores and results	high	Pass: After a score is submitted, the system recalculates standings and applies the configured tiebreaker rules when teams have equal rankings.	Keeps standings accurate and up to date.
NFR-001	Non-Functional	The system shall recalculate and update tournament standings, goal differentials, tiebreakers, and the leaderboard within 200 ms of a valid score submission	high	Pass: Performance testing shows updates are completed within 200 ms after score submission. Fail: Updates take more than 200 ms.	Ensures that live tournament information remains up to date.
NFR-002	Non-Functional	The system shall preserve submitted match scores and tournament data without data loss during normal system operation.	high	Pass: After successfully submitting a match score, refreshing or reopening the system displays the same saved score and updated tournament state. Fail: Submitted tournament data is missing or altered without authorization.	Ensures reliable tournament management and prevents loss of important match results.



Use-Case Flow Specification

Use Case: Record / Update Match Score

Use Case ID: UC-04

Primary Actor: Tournament Director

Preconditions

1. The Tournament Director is authorized to manage the tournament.
2. The tournament has already been created.
3. Teams have been added and the tournament bracket has been generated.
4. The selected match exists and is available for score entry.

Postconditions

1. The match score is successfully saved.
2. The match result is updated.
3. The winning team is advanced to the appropriate next match in the tournament bracket.
4. Tournament standings and tiebreakers are recalculated and updated.

Main Success Scenario

1. The Tournament Director selects a tournament.
2. The system displays the tournament bracket and available matches.
3. The Tournament Director selects a completed or live match.
4. The system displays the selected match and the participating teams.
5. The Tournament Director enters or updates the scores for both teams.
6. The Tournament Director submits the match score.
7. The system validates the entered scores.
8. The system saves the match score and determines the match result.
9. The system identifies the winning team.
10. The system automatically advances the winning team to the appropriate next match in the bracket.
11. The system recalculates the tournament standings, goal differentials, and tiebreakers.
12. The system updates the leaderboard and tournament information.
13. The system displays a confirmation message showing that the match result has been successfully updated.
14. The use case ends successfully.

Alternate Flow: Invalid or Incomplete Score

7a. Invalid or incomplete score entered

1. The system detects that one or more score values are missing or invalid.
2. The system does not save the match result.
3. The system displays an error message indicating that valid scores must be entered.
4. The Tournament Director corrects the score values.
5. The flow returns to Step 6 of the Main Success Scenario.